

Privacy with Asymmetrically Informed Consumers

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CPSC 6551 Final Project

We don't know who uses our data for what



Health Care

Health Insurers Are Vacuuming Up Details About You — And It Could Raise Your Rates

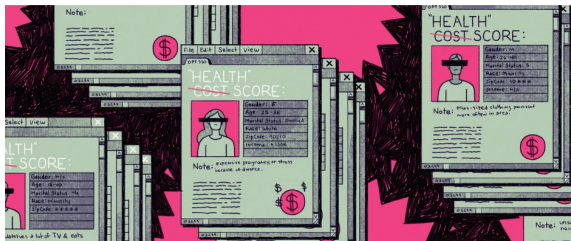


Figure: A 2018 ProPublica report

ProPublica article

With little public scrutiny, the health insurance industry has joined forces with data brokers to vacuum up personal details about hundreds of millions of Americans, including, odds are, many readers of this story. The companies are tracking your race, education level, TV habits, marital status, net worth. They're collecting what you post on social media, whether you're behind on your bills, what you order online. Then they feed this information into complicated computer algorithms that spit out predictions about how much your health care could cost them.

Figure: Worried about price discrimination or denying coverage. The Affordable Care Act prohibits some of this.

Overview

Question

What happens when consumers don't know what their data will be used for?

Plan:

- Build off a classic theory paper (Taylor 2004)
 - Review results about sophisticated and naive consumers
 - Consider some extensions

Previous literature on the economics of privacy

- **Reviews:** Acquisti, Taylor, Wagman (2016), Acquisti (2023), Acquisti, Brandimarte, Loewenstein (2015)
 - Empirical work shows how consumers don't understand what will happen to data.
- **Naive consumers:** *Taylor (2004)*, Acquisti and Varian (2005), Bisceglia, Bonatti, Scott Morton (2025), Heidhues and Köszegi (2017)
 - Assumes that consumers either perfectly foresee the harms of revealing data or are fully naive.
- **Dynamic pricing and price discrimination:** *Taylor (2004)*, Acquisti and Varian (2005), Ichihashi (2020)

Taylor (2004)

RAND Journal of Economics
Vol. 35, No. 4, Winter 2004
pp. 631–650

Consumer privacy and the market for customer information

Curtis R. Taylor*

Figure: Taylor (2004) in the *RAND Journal of Economics*

Setup

- Two periods
- Mass one of consumers, indexed by $i \in [0, 1]$
 - Risk-neutral, do not discount the future
- In each period $t = 1, 2$, each consumer demands one unit of good t

Consumer valuations

- Each consumer i 's valuation in period t is $v_{i,t} \in \{v_L, v_H\}$,
 $v_H > v_L > 0$
- Random variables v_{i1}, v_{i2} i.i.d.

$$\Pr[v_{it} = v_H | \lambda_i] = \lambda_i$$

- $\lambda_i \in [0, 1]$ i.i.d. in population. Drawn from CDF $F(\lambda)$.
- λ not observed, $F(\lambda)$ commonly known

Firms

- Firm t is monopolist seller of good t
 - No production costs
 - Risk neutral, do not discount future
- Firm 1 can sell customer list to firm 2
 - Customer list is $q_{i1} \in \{0, 1\}$ for all $i \in [0, 1]$

Timing

- 1 Consumer privately observes v_{i1}, v_{i2}
- 2 Firm 1 posts price $p_1 \in \mathbb{R}_+$.
 - Consumer buys ($q_{i1} = 1$) or rejects ($q_{i1} = 0$)
 - Firm 2 observes the offer.
- 3 In “disclosure” regime, firm 1 may offers customer list for price $w \in \mathbb{R}_+$.
 - Firm 2 may buy ($x = 1$) or reject ($x = 0$) list.
 - In “confidential” regime, firm 1 cannot sell data.
- 4 With list, firm 2 offers $p_{i2} = p_2^1$ to buyers and $p_{i2} = p_2^0$ to refusers. Without list, firm 2 offers $p_{i2} = p_2$.
 - Consumers buy ($q_{i2} = 1$) or reject ($q_{i2} = 0$).

Payoffs

- Consumer i payoff:

$$(v_{i1} - p_1)q_{i1} + (v_{i2} - p_2)q_{i2}$$

- Firm payoffs under “confidential regime”:

- Firm 1: $p_1 Q_1$, where Q_1 is the mass of buyers
- Firm 2: $p_2 Q_2$

- Firm payoffs under “disclosure regime”:

- Firm 1: $p_1 Q_1 + wx$
- Firm 2: $(1 - x)p_2 Q_2 + x(p_2^1 Q_2^1 + p_2^0 Q_2^0 - w)$, where Q_2^1, Q_2^0 are the masses of consumers who buy at p_2^1, p_2^0 respectively

Solution concept

- Efficient Perfect Bayesian Equilibrium
 - PBE where ties are resolved in favour of social efficiency.

Pricing in the confidential regime

Proposition 1

There is a unique efficient PBE under the confidential regime.
Equilibrium strategies are

$$p_t = \begin{cases} v_L, & \text{if } \mathbb{E}[\lambda] \leq v_L/v_H \\ v_H, & \text{if } \mathbb{E}[\lambda] > v_L/v_H \end{cases}$$

$$q_{it} = \begin{cases} 1, & \text{if } p_t \leq v_{it} \\ 0, & \text{if } p_t > v_{it} \end{cases}$$

for any $i \in [0, 1]$ and $t = 1, 2$.

Intuition for optimal pricing

- $\Pr[v_{it} = v_H] = \mathbb{E}[\Pr[v_{it} = v_H | \lambda_i]] = \mathbb{E}[\lambda]$
- Suppose $v_H = 2, v_L = 1$.
- Then $p = v_H$ only when $\Pr[v_{it} = v_H] > 1/2$

$$\Leftrightarrow \mathbb{E}[\lambda] > v_L/v_H$$

- When demand is “inelastic”

Confidential Regime

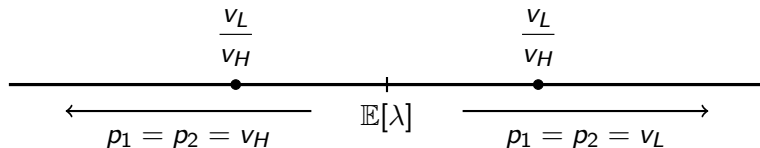


Figure: Equilibrium (Proposition 1)

Optimal pricing in general

Lemma 1

In any PBE, firm 2's pricing strategy must satisfy

$$p_{i2} = \begin{cases} v_L, & \text{if } \Pr[v_{i2} = v_H | k_i] < v_L/v_H \\ v_H, & \text{if } \Pr[v_{i2} = v_H | k_i] > v_L/v_H \end{cases}$$

for any $i \in [0, 1]$, where k_i is what firm 2 knows about i .

Disclosing data with naive consumers

Proposition 2

There is a unique efficient PBE outcome of the game, as follows.

- 1 For some $\bar{\gamma} > 1$, if $\bar{\gamma}\mathbb{E}[\lambda] \leq v_L/v_H$
 - $p_1 = p_2^1 = v_L$. Price of customer list is zero. All customers buy both goods.
- 2 If $v_L/v_H < \bar{\gamma}\mathbb{E}[\lambda]$
 - Firm 1 charges $p_1 = v_H$. Consumer i buys iff $v_{i1} = v_H$
 - Price of the customer list is positive
 - Firm 2 charges $p_2^1 = v_H$ to buyers and $p_2^0 = v_L$ to refusers. Only (v_H, v_L) consumers do not buy good 2.

Naive consumers: visualizing Proposition 2

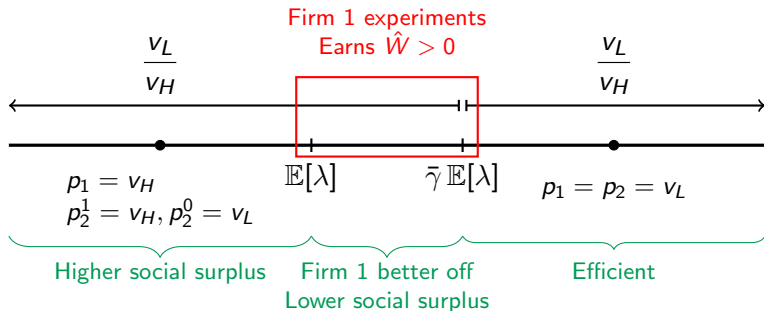


Figure: Black for equilibrium (Proposition 2).
Green for welfare (Corollary 1).

Naivety and perfect Bayesian equilibrium

- NB: naivety doesn't seem compatible with PBE?
- When $\mathbb{E}[\lambda] < v_L/v_H \leq \bar{\gamma}\mathbb{E}[\lambda]$
 - Consumers believe their data is kept private
 - Consumers know firm 1 should set $p_1 = v_L$ when $\mathbb{E}[\lambda] < v_L/v_H$
 - But firm 1 sets $p_1 = v_H$, so they must be up to no good... like selling data

Strategies for sophisticated consumers

Lemma A4

Consider the continuation game that begins after firm 1 sets p_1 .

- 1 If $p_1 = v_L$, then there exists a PBE of the continuation game in which all consumers purchase good 1.
- 2 If $p_1 = v_H$, then there exists a PBE of the continuation game where:
 - 1 All type- (v_L, v_L) consumers refuse to purchase good 1.
 - 2 All (v_L, v_H) refuse.
 - 3 All (v_H, v_L) buy.
 - 4 Fraction ρ^* of (v_H, v_H) consumers refuse.
- No other PBE of the respective continuation games yields a higher expected payoff to firm 1.

(v_H, v_H) make the customer list worthless!

Some intuition:

- Suppose there were 4 (v_L, v_L) types and 6 (v_H, v_H) types

- Not an equilibrium:

Refusers: 4 (v_L, v_L) , 3 (v_H, v_H)

Buyers: 3 (v_H, v_H)

- Firm 2 would charge $p_2^0 = v_L$, $p_2^1 = v_H$, so (v_H, v_H) would want to deviate (refuse)

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- Not an equilibrium:
Refusers: 4 (v_L, v_L) , 3 (v_H, v_H)
Buyers: 3 (v_H, v_H)
 - Firm 2 would charge $p_2^0 = v_L$, $p_2^1 = v_H$, so (v_H, v_H) would want to deviate (refuse)
- Equilibrium:
Refusers: 4 (v_L, v_L) , 4 (v_H, v_H)
Buyers: 2 (v_H, v_H)
 - Firm 2 would charge $p_2^0 = p_2^1 = v_H$
- ρ^* of (v_H, v_H) reject, ρ^* calibrated to force firm 2 to charge the same prices.

Equilibrium: disclosure, sophisticated consumers

Proposition 4

There is a unique efficient PBE outcome as follows.

- 1 If $\mathbb{E}[\lambda] \leq v_L/v_H$, then firm 1 charges $p_1 = v_L$
 - All consumers purchase good 1 and 2
 - Price of customer list $\hat{W} = 0$, $p_2 = v_L$
- 2 For some $\underline{\gamma} < 1$, if $\underline{\gamma}\mathbb{E}[\lambda] \leq v_L/v_H < \mathbb{E}[\lambda]$, then
 - Firm 1 charges $p_1 = v_L$ and all consumers buy.
 - Price of customer list $\hat{W} = 0$, $p_2^1 = v_H$. Consumer i buys iff $v_{i2} = v_H$.
- 3 If $v_L/v_H < \underline{\gamma}\mathbb{E}[\lambda]$
 - Firm 1 charges $p_1 = v_H$. All (v_H, v_L) and $(1 - \rho^*)$ of (v_H, v_H) consumers buy.
 - Price of customer list is zero, $p_2^1 = p_2^0 = v_H$. Consumer i buys iff $v_{i2} = v_H$.

Disclosing data with sophisticated consumers

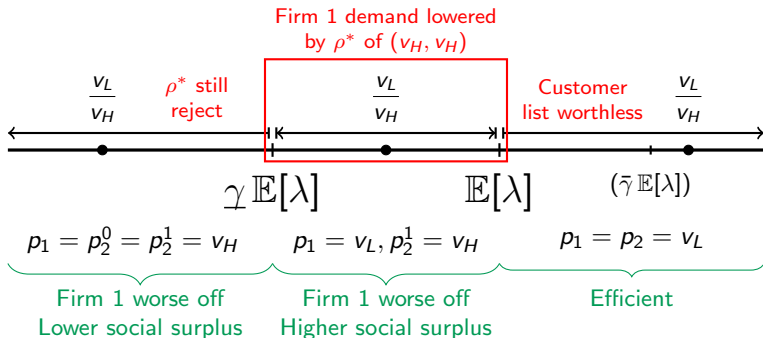


Figure: Black for equilibrium (Proposition 4).
Green for welfare (Corollary 2).

Firm 1 would like to commit to privacy

- Firm 1 is worse off in the disclosure-sophisticated regime than in the confidential regime
- Firm 1 would like to commit to not selling data if possible.
 - Typical result in this literature.

Extension 1: consumers have intermediate beliefs

Definition

All consumers believe firm 2 is sold the customer list with probability θ .

Remark

Without loss of generality, firm 1 sells the customer list to firm 2 with probability 1.

Consumers still protect their privacy

Claim

For $\theta \in (0, 1)$, the strategies and beliefs from Proposition 4 are still a PBE.

- If $p_1 = v_H$, then ρ^* of (v_H, v_H) strategically refuse good 1.

...because refusing good 1 is costless

Proof sketch

- At $p_1 = v_H$, type (v_H, v_H) are indifferent between buying and not buying.
- Whether they believe firm 1 sells data or not, ρ^* rejecting is optimal.
- Firm beliefs and best responses are as before. We have the same PBE for all $\theta \in (0, 1)$.



Extension 2: data markets

- Suppose firm 1 no longer sells a good
- Rather, consumers sell v_{i1} to firm 1 for price p_1 .
- Rest of game proceeds as before with fully sophisticated consumers.

Nowhere for high types to hide

Conjecture

All types sell their data at price zero.

Idea

- (v_L, v_L) will sell their data at price zero, so (v_L, v_H) will too.
- If $(v_H, \cdot)$ don't sell, they will reveal themselves as high types.
- So (v_H, v_L) and (v_H, v_H) may as well sell anyway. They can only sell at price zero.